Z10.5 (cont)

NOTES TO REPORT TM2-DSBC(CSR) (i)

1. This report is to be used for recording the thickness measurement of:-

   Strength deck plating and sheerstrake plating transverse sections:-

   One, two or three sections within the cargo length area, comprising of the structural items (1), (2) and (3) as shown on the diagram of typical transverse sections illustrated on sheets 11 and 12 of this document.

2. Only the deck plating strakes outside the line of openings are to be recorded.

3. The topside area comprises deck plating, stringer plate and sheerstrake (including rounded gunwales).

4. The exact frame station of measurement is to be stated.

5. The single measurements recorded are to represent the average of multiple measurements.

6. The remaining corrosion addition is to be recorded with result of gauged thickness minus renewal thickness. If the result is negative, the structure in way shall be renewed, and the mark “R” is to be indicated in the right-hand column. If the result is between 0 and 0.5 mm (0 included), the structure in way shall be additional gauged, and the mark “S” is to be indicated in the right-hand column.

Z10.5 (cont) TM2-DSBC(CSR) (ii) Report on THICKNESS MEASUREMENT OF SHELL PLATING (one, two or three transverse sections)

Sheet 6

Ship's name..... Class Identity No..... Report No.....

SHELL PLATING																																
	FIRST TRANSVERSE SECTION AT FRAME NUMBER										SECOND TRANSVERSE SECTION AT FRAME NUMBER										THIRD TRANSVERSE SECTION AT FRAME NUMBER									
STRAKE POSITION	No. or Letter	As Built Thk. mm	Vol. Thk. Add. mm	Ren. Thk. mm (a)	Gauged Thk. mm (b)		Remaining Corr. Addition, mm (b)-(a)				No. or Letter	As Built Thk. mm	Vol. Thk. Add. mm	Ren. Thk. mm (a)	Gauged Thk. mm (b)		Remaining Corr. Addition, mm (b)-(a)				No. or Letter	As Built Thk. mm	Vol. Thk. Add. mm	Ren. Thk. mm (a)	Gauged Thk. mm (b)		Remaining Corr. Addition, mm (b)-(a)			
					P	S	P		S						P	S	P		S						P	S	P		S	
1st below sheer strake																														
2nd																														
3rd																														
4th																														
5th																														
6th																														
7th																														
8th																														
9th																														
10th																														
11th																														
12th																														
13th																														
14th																														
15th																														
16th																														
17th																														
18th																														
19th																														
20th																														
Keel strake																														
BOTTOM TOTAL																														

Operators Signature..... NOTES – See Reverse

Z10.5 (cont)

NOTES TO REPORT TM2-DSBC(CSR) (ii)

1. This report is to be used for recording the thickness measurement of:-

   Shell plating at transverse sections:-

   One, two or three sections within the cargo length area, comprising of the structural items (4), (5), (6) and (7) as shown on the diagram of typical transverse sections illustrated on sheets 11 and 12 of this document.

2. The bottom area comprises keel, bottom and bilge plating.

3. The exact frame station of measurement is to be stated.

4. The single measurements recorded are to represent the average of multiple measurements.

5. The remaining corrosion addition is to be recorded with result of gauged thickness minus renewal thickness. If the result is negative, the structure in way shall be renewed, and the mark “R” is to be indicated in the right-hand column. If the result is between 0 and 0.5 mm (0 included), the structure in way shall be additional gauged, and the mark “S” is to be indicated in the right-hand column.

Z10.5 (cont) **TM3-DSBC(CSR)**

Report on THICKNESS MEASUREMENT OF LONGITUDINAL MEMBERS (one, two or three transverse sections)

Sheet 7

Ship's name..... Class Identity No..... Report No.....

	FIRST TRANSVERSE SECTION AT FRAME NUMBER										SECOND TRANSVERSE SECTION AT FRAME NUMBER										THIRD TRANSVERSE SECTION AT FRAME NUMBER									
STRUCTURAL MEMBER	Item No.	As Built Thk. mm	Vol. Thk. Add. mm	Ren. Thk. mm (a)	Gauged Thk. mm (b)		Remaining Corr. Addition, mm (b)-(a)				Item No.	As Built Thk. mm	Vol. Thk. Add. mm	Ren. Thk. mm (a)	Gauged Thk. mm (b)		Remaining Corr. Addition, mm (b)-(a)				Item No.	As Built Thk. mm	Vol. Thk. Add. mm	Ren. Thk. mm (a)	Gauged Thk. mm (b)		Remaining Corr. Addition, mm (b)-(a)			
					P	S	P		S						P	S	P		S						P	S	P		S	

Operators Signature..... NOTES – See Reverse

Z10.5 (cont)

NOTES TO REPORT TM3-DSBC(CSR)

1. This report is to be used for recording the thickness measurement of:-

   Longitudinal Members at transverse sections:-

   One, two or three sections within the cargo length area, comprising of the appropriate structural items (8) to (20) and (31) as shown on diagram of typical transverse sections illustrated on sheets 11 and 12 of this document.

2. The exact frame station of measurement is to be stated.

3. The single measurements recorded are to represent the average of multiple measurements.

4. The remaining corrosion addition is to be recorded with result of gauged thickness minus renewal thickness. If the result is negative, the structure in way shall be renewed, and the mark “R” is to be indicated in the right-hand column. If the result is between 0 and 0.5 mm (0 included), the structure in way shall be additional gauged, and the mark “S” is to be indicated in the right-hand column.

Z10.5 (cont) **TM4-DSBC(CSR)**

Report on THICKNESS MEASUREMENT OF TRANSVERSE STRUCTURAL MEMBERS
In the double bottom, hopper side and topside water ballast tanks

Sheet 8

Ship's name..... Class Identity No..... Report No.....

TANK DESCRIPTION:

LOCATION OF STRUCTURE

STRUCTURAL MEMBER	ITEM	As Built Thickness mm	Voluntary Thickness Addition mm	Renewal Thickness mm (a)	Gauged Thickness mm (b)		Remaining Corr. Addition mm (b)-(a)			
					P	S	P		S	

Operators Signature..... NOTES – See Reverse

Z10.5 (cont)

NOTES TO REPORT TM4-DSBC(CSR)

1. This report is to be used for recording the thickness measurement of:-

   Transverse structural members, comprising of the appropriate structural items (23) to (25) and (34) as shown on diagram of typical transverse section, illustrated on sheets 11 and 12 of this document.

2. Guidance for areas if measurement is indicated on the diagrams shown on sheet 13 of this document.

3. The single measurements recorded are to represent the average of multiple measurements.

4. The remaining corrosion addition is to be recorded with result of gauged thickness minus renewal thickness. If the result is negative, the structure in way shall be renewed, and the mark “R” is to be indicated in the right-hand column. If the result is between 0 and 0.5 mm (0 included), the structure in way shall be additional gauged, and the mark “S” is to be indicated in the right-hand column.

Z10.5 (cont) **TM5-DSBC(CSR)** **Report on THICKNESS OF CARGO HOLD TRANSVERSE BULKHEADS**

Sheet 9

Ship's name..... Class Identity No..... Report No.....

HOLD DESCRIPTION:

LOCATION OF STRUCTURE: FRAME NO.:

STRUCTURAL COMPONENT (PLATING/STIFFENER)	As Built Thickness mm	Voluntary Thickness Addition mm	Renewal Thickness mm (a)	Gauged Thickness mm (b)		Remaining Corr. Addition mm (b)-(a)			
				P	S	P		S	

Operators Signature..... NOTES – See Reverse

Z10.5 (cont)

NOTES TO REPORT TM5-DSBC(CSR)

1. This report form is to be used for recording the thickness measurement of:-

   W.T. transverse bulkheads in cargo holds

2. Guidance for areas of measurement is indicated on the diagrams shown on sheet 13 of this document.

3. The single measurements recorded are to represent the average of multiple measurements.

4. The remaining corrosion addition is to be recorded with result of gauged thickness minus renewal thickness. If the result is negative, the structure in way shall be renewed, and the mark “R” is to be indicated in the right-hand column. If the result is between 0 and 0.5 mm (0 included), the structure in way shall be additional gauged, and the mark “S” is to be indicated in the right-hand column.

Z10.5 (cont) **TM6-DSBC(CSR)** **Report on THICKNESS MEASUREMENT OF MISCELLANEOUS STRUCTURAL MEMBERS**

Sheet 10

Ship's name..... Class Identity No..... Report No.....

STRUCTURAL MEMBER:										SKETCH
LOCATION OF STRUCTURE:										
Description	As Built Thk. mm	Voluntary Thickness Addition mm	Renewal Thickness mm (a)	Gauged Thickness mm (b)		Remaining Corr. Addition mm (b)-(a)				
				P	S	P		S		

Operators Signature..... NOTES – See Reverse

Z10.5 (cont)

NOTES TO REPORT TM6-DSBC(CSR)

1. This report is to be used for recording the thickness measurement of:-

   Miscellaneous structural members including the structural items (28) as shown on diagram of typical transverse section illustrated on sheets 11 and 12 of this document.

2. Guidance for areas of measurement is indicated on sheet 13 of this document.

3. The single measurements recorded are to represent the average of multiple measurements.

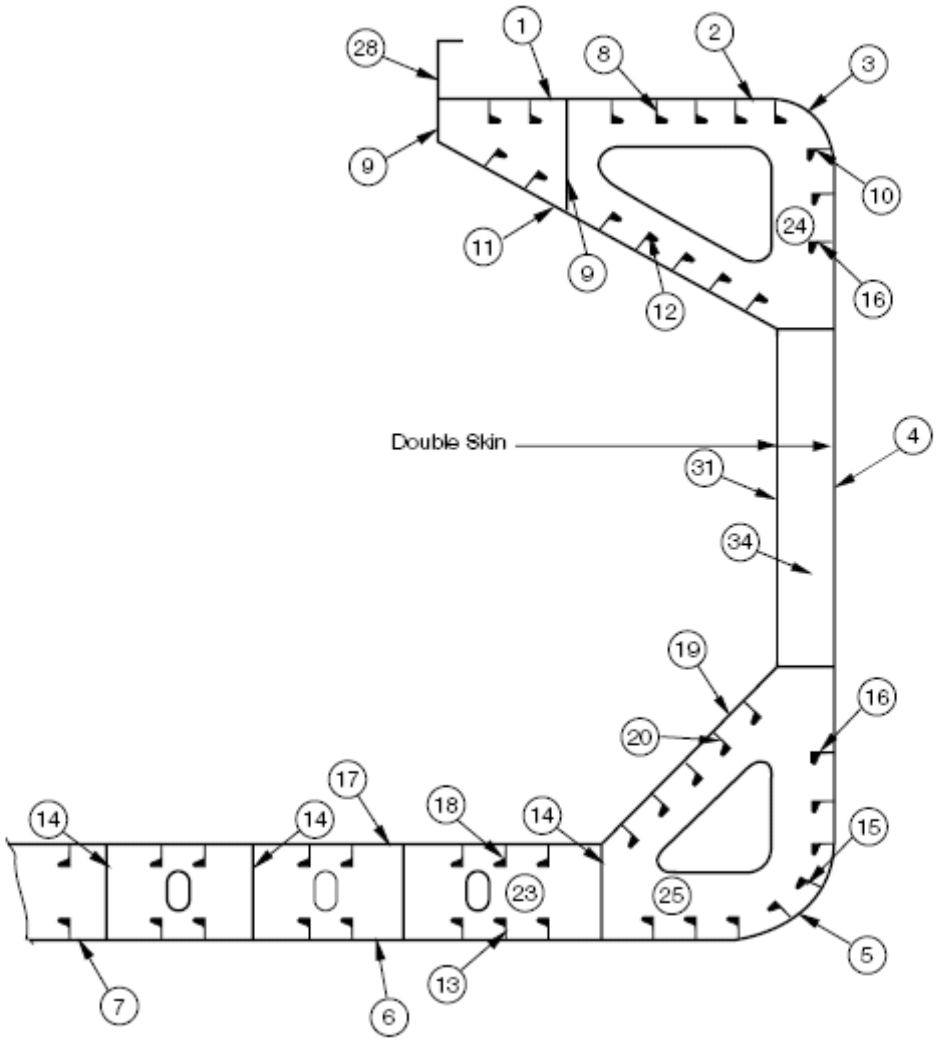
4. The remaining corrosion addition is to be recorded with result of gauged thickness minus renewal thickness. If the result is negative, the structure in way shall be renewed, and the mark “R” is to be indicated in the right-hand column. If the result is between 0 and 0.5 mm (0 included), the structure in way shall be additional gauged, and the mark “S” is to be indicated in the right-hand column.

Z10.5
(cont)

Sheet 11

Thickness Measurement - Double Skin Bulk Carriers

Typical transverse section of a double skin bulk carrier with indication of longitudinal and transverse members



Report on TM2-DSBC(CSR) (i) & (ii)	
1.	Strength deck plating
2.	Stringer plate
3.	Sheerstrake
4.	Side shell plating
5.	Bilge plating
6.	Bottom plating
7.	Keel plate

Report on TM3-DSBC(CSR)			
8.	Deck longitudinals	17.	Inner bottom plating
9.	Deck girders	18.	Inner bottom longitudinals
10.	Sheerstrake longitudinals	19.	Hopper plating
11.	Topside tank sloping plate	20.	Hopper longitudinals
12.	Topside tank sloping plate longitudinals	31.	Inner side plating
13.	Bottom longitudinals	-	Inner side longitudinals, if any
14.	Bottom girders	-	Horizontal girders in wing ballast tanks
15.	Bilge longitudinals		
16.	Side shell longitudinals, if any		

Report on TM4-DSBC(CSR)	
23.	Double bottom tank floors
24.	Topside tank transverses
25.	Hopper side tank transverses
34.	Transverse web frame
-	Ordinary transverse frame in double skin tank

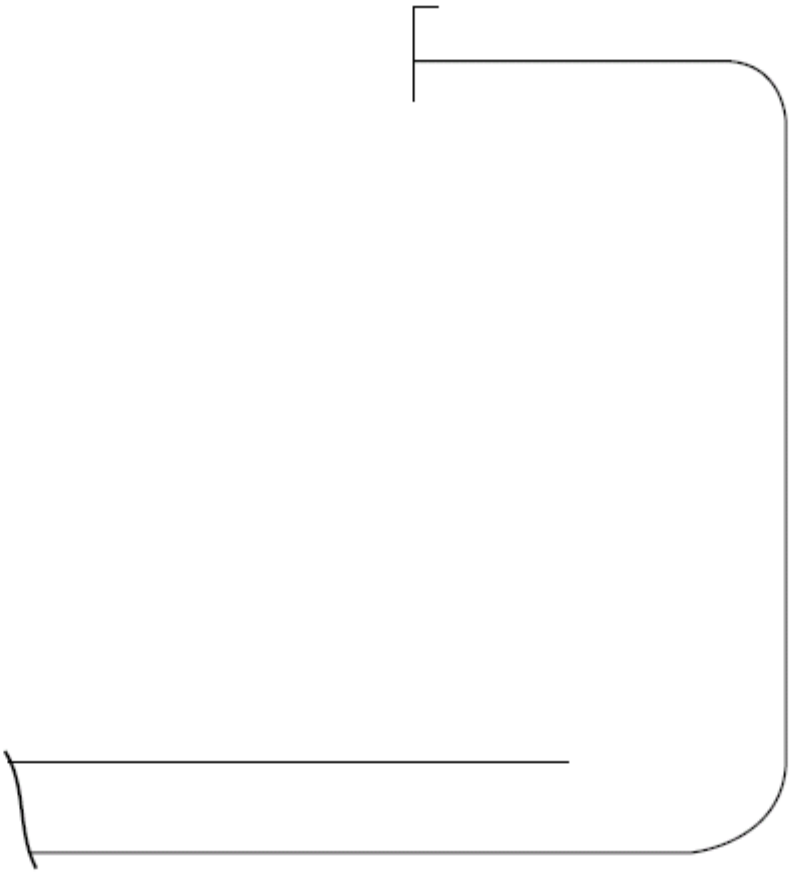
Report on TM6-DSBC(CSR)	
28.	Hatch coamings
-	Deck plating between hatches
-	Hatch covers

Z10.5 (cont)

Thickness Measurement - Double Skin Bulk Carriers

Sheet 12

Transverse section outline: This diagram may be used for those ships where the diagram on sheet 11 is not suitable



Report on TM2-DSBC(CSR) (i) & (ii)	
1.	Strength deck plating
2.	Stringer plate
3.	Sheerstrake
4.	Side shell plating
5.	Bilge plating
6.	Bottom plating
7.	Keel plate

Report on TM3-DSBC(CSR)			
8.	Deck longitudinal	17.	Inner bottom plating
9.	Deck girders	18.	Inner bottom longitudinals
10.	Sheerstrake longitudinals	19.	Hopper plating
11.	Topside tank sloping plating	20.	Hopper longitudinals
12.	Topside tank sloping plating longitudinal	31.	Inner side plating
13.	Bottom longitudinals	-	Inner side longitudinals, if any
14.	Bottom girders	-	Horizontal girders in wing ballast tanks
15.	Bilge longitudinals		
16.	Side shell longitudinals, if any		

Report on TM4-DSBC(CSR)	
23.	Double bottom tank floors
24.	Topside tank transverses
25.	Hopper side tank transverses
34.	Transverse web frame
-	Ordinary transverse frame in double skin tank

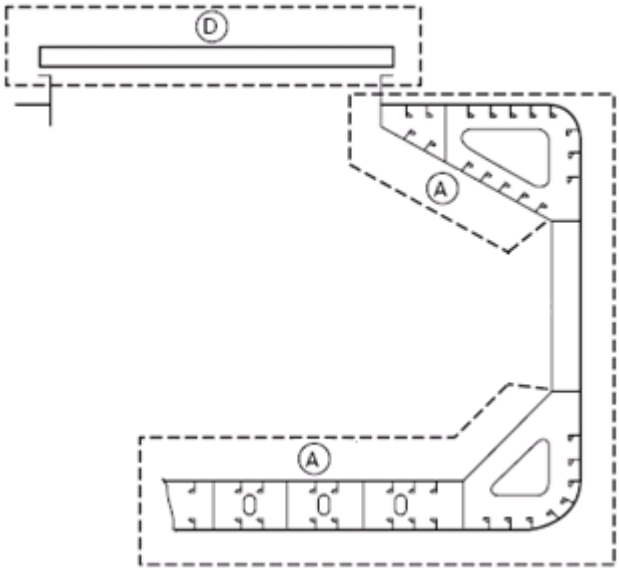
Report on TM6-DSBC(CSR)	
28.	Hatch coamings
-	Deck plating between hatches
-	Hatch covers

Z10.5
(cont)

Sheet 13

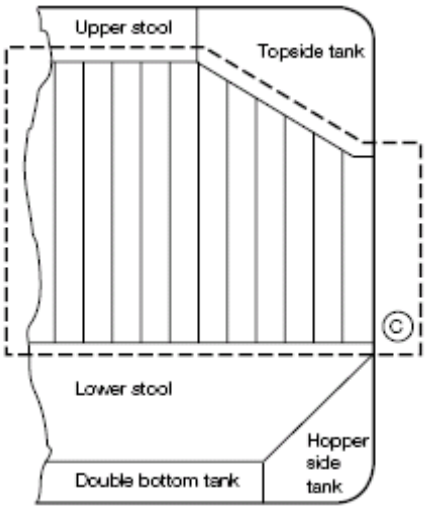
Close-up Survey and Thickness Measurement Areas

Typical transverse section
Areas A and D



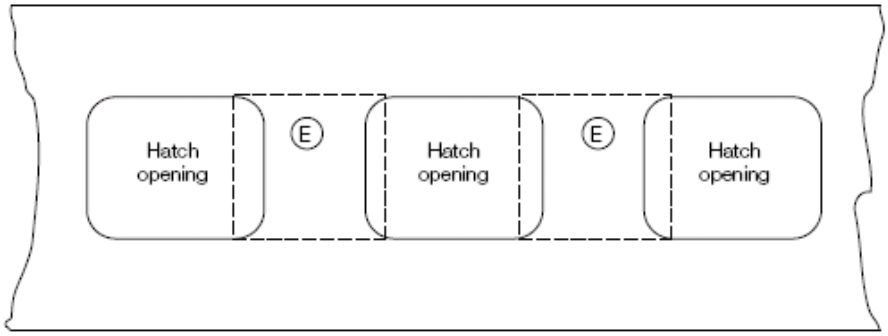
Thickness to be reported on TM3-DSBC(CSR), TM4-DSBC(CSR),TM6-DSBC(CSR) as appropriate

A cargo hold, transverse bulkhead
Area C



Thickness to be reported on TM5-DSBC(CSR)

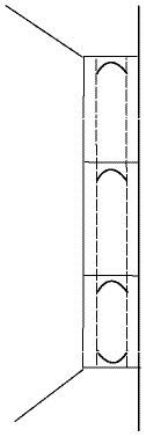
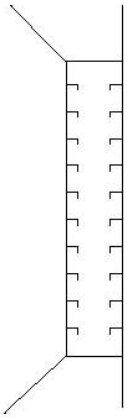
Typical areas of deck plating inside line
of hatch openings between cargo hold hatches
Area E



Thickness to be reported on TM6-DSBC(CSR)

Z10.5 (cont)

Framing in double-side tanks
Area (B)

	
Ordinary transverse frame in double skin tank	Ordinary longitudinal structure in double skin tank

Thickness to be reported on TM4-DSBC(CSR)

End of Annex II

End of Annex II